

ABC E-COMMERCE

Sales Data Cleaning, Analysis & Dashboard Report

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1. Executive Summary

ABC E-Commerce is an online retail company operating across multiple regions. Management identified recurring inconsistencies in monthly sales reports, traced to underlying data quality issues in the source dataset of 5,000 sales transactions. This report documents the cleaning process applied to the raw data, presents the resulting sales performance analysis, and describes the interactive Excel dashboard built to support ongoing business decision-making.

2. Data Quality Issues Identified

The raw dataset contained several common e-commerce data problems:

- **Missing values** gaps in customer rating, delivery days, and payment method fields.
- **Duplicate records** repeated transaction rows inflating totals.
- **Inconsistent formatting** mixed casing/whitespace fragmenting categories.
- **Invalid entries** negative prices, impossible delivery days, or ratings outside 1–5.

3. Data Cleaning Methodology

Using Excel Power Query, the following steps were applied:

- Imputation of missing numeric values with medians or category averages; text gaps replaced with “Unknown.”
- Removal of duplicates via transaction IDs.
- Standardization of category/region labels.
- Correction of invalid entries against business rules.

4.Key Performance Indicators

Total Revenue	\$5,104,379.37
Total Quantity	20,201
Average rating	2.97/5
Average delivery days	6.12
Total Orders	5,000

5.Revenue Breakdown

By Category

PRODUCT CATEGORY	REVENUE\$
Electronics	\$1,829,297.34
Clothing	\$1,531,572.32
Home	\$966,601.74
Beauty	\$761,425.79

By Region

REGION	REVENUE\$
West	\$1,345,561.03
North	\$1,281,508.45
South	\$1,242,208.81
East	\$1,235,104.08

6. Payment Method Distribution

METHOD	SHARE OF ORDERS
Card	46%
Cash on delivery	35%
Wallet	19%

7. Business Insights

- Electronics is the highest revenue generator prioritize inventory and marketing.
- West region leads performance replicate strategies elsewhere.
- Card payments dominate opportunity to grow wallet usage with incentives.
- Delivery times (6.11 days) plus low ratings (2.97/5) fulfilment or customer experience needs urgent improvement.

8. Recommendations

- Investigate recent revenue decline to confirm trend vs incomplete data.
- Launch wallet incentive campaigns to reduce COD reliance.
- Review logistics/courier performance to shorten delivery times.
- Continue prioritizing Electronics and Clothing while reassessing Beauty and Home demand.

9. Conclusion

The cleaning process resolved missing values, duplicates, inconsistent formatting, and invalid entries, producing a reliable dataset. The dashboard now provides management with clear visibility into performance by category, payment method, and region. Key insights highlight

the strength of Electronics and the West region, the dominance of card payments, and the urgent need to improve delivery speed and customer satisfaction.